

BHARAT SREYUS ANANDHAVALLI VETRIVEL

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EDUCATION

Master of Computer Science

Aug 2026 - May 2028

University of Illinois, Urbana-Champaign, United States

Bachelor of Engineering in Computer Science and Engineering

Aug 2018 - Jun 2022

Anna University, India

CGPA: 8.98/10.00

TECHNICAL SKILLS

Languages	Java, Python, SQL, JavaScript, C++
Frameworks	Agent Development Kit (ADK), A2A Protocol, Model Context Protocol (MCP)
AI / Deep Learning / ML	WVCS, RAG, scikit-learn, ONNX, SBERT, CNNs, LSTM, TF, PyTorch, MediaPipe
Databases	PostgreSQL, MSSQL, PGVector
Tools	Git, Mercurial (Hg), IntelliJ, VSCode, GitHub Copilot, Claude Code, CursorAI

EXPERIENCE

MEMBER TECHNICAL STAFF (SOFTWARE ENGINEER), ZOHU CORPORATION

Jun 2022 - Jun 2026

ManageEngine ServiceDesk Plus (On-Premise), Chennai, India

- Shipped ME-MCP-v1, a bidirectional MCP framework in Java that both serves and consumes tools — 60+ internal ServiceDesk Plus REST APIs exposed to any MCP client, adopted by 6 product teams across ManageEngine's Suite — by implementing MCP server, client, and tool hosting natively, letting Zia bot invoke third-party MCP servers' tools.
- Extended the framework into ME-MCP-v2 for cross-product agent interoperability — adopted by 2 product teams — by embedding Google's Agent Development Kit (ADK) alongside the MCP tooling, porting ADK's session and state design onto Zoho's Mickey DB framework in Java, and adding Agent2Agent (A2A) protocol hosting.
- Delivered intelligent solution suggestion in Zia bot — cutting ticket-creation flow by 60% — by replacing Lucene-based indexing with a RAG pipeline over internal KB articles and user queries.
- Scaled multilingual approval prediction for ServiceDesk Plus On-Premise — from 3 to 43 supported languages and 3 to 4 prediction labels — by retraining the full model for a new "Need More Clarification" class and reworking the signature-removal preprocessing pipeline.
- Automated end-to-end issue resolution in the issue-management system — reducing manual effort by ~50% — by designing a multi-agent pipeline on GitHub Copilot with custom AGENTS.md instructions spanning bug analysis, fix generation, and test-case synthesis.

PROJECT INTERN, ZOHU CORPORATION, Chennai, India

Feb 2022 - May 2022

- Delivered feature rollouts on schedule under Agile methodology — by owning rollout coordination from sprint planning through release sign-off.
- Expanded automated validation coverage for the QA team — by defining comprehensive test cases and automating validation workflows in Python.
- Resolved customer-reported product issues — by conducting support-ticket analysis with support technicians and delivering root cause analyses (RCA) and fix jars.

RESEARCH INTERN, SSN INSTITUTIONS, Chennai, India

Jun 2021 - Jul 2021

- Automated blood-cell mask generation on the ErythrocytesIDB database — saving 1–2 hours of manual image segmentation — by training a U-Net deep-learning segmentation model in Python.
- Built the image pre-processing pipeline in Python and OpenCV, covering normalization, masking, and dataset preparation, to produce training-ready inputs for the segmentation model.

ACADEMIC PROJECTS

PARKINSON'S DISEASE DETECTION (UG Final Year Project)

Oct 2021 - Apr 2022

- Detected Parkinsonian hand tremors from a live camera feed in a Python web application — matching against a 550-patient PDD dataset at a 0.67 cosine-similarity threshold — by tracking hand landmarks with Google's MediaPipe and OpenCV, then comparing the captured tremor pattern to that of a PD patient by cosine similarity.

CONVERSATIONAL CHATBOT USING REDDIT DATA

May 2021 - Jun 2021

- Built a conversational chatbot in Python using Google's Neural Machine Translation (NMT) model, trained on Reddit conversational data.

ACHIEVEMENTS AND EXTRA-CURRICULARS

- Youngest recipient of the Zenith (Top Performer) rating at the product level, recognizing exceptional technical and leadership contributions.
- Active volunteer at Tamil Nadu Blind Football Association (TNBFA) coaching partially blind football players.